

CARDSTON, AB, Canada

WMO: 711530

Lat: 49.201N

Lon: 113.286W

Elev: 1136

StdP: 88.40

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-24.1	-30.9	0.2	-27.2	-27.2	0.4	-23.6	12.7	6.5	11.5	5.7	2.0	220	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	30.8	16.3	28.9	15.8	27.0	15.2	18.3	26.2	17.2	25.5	16.3	24.7	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	12.9	21.5	14.4	11.8	20.0	13.1	10.8	18.8	56.6	26.1	52.9	25.5	49.8	24.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	8.0	DB	-32.7	33.9	3.8	2.1	-35.4	35.4	-37.7	36.6	-39.8	37.8	-42.6	39.3	(4)
(5)			WB	-32.8	20.7	3.8	1.4	-35.6	21.7	-37.8	22.6	-39.9	23.4	-42.6	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-4.5	-4.1	-0.4	4.7	9.9	13.8	18.0	17.3	12.4	6.0	-0.1	-4.5	(6)	
	DBStd	10.19	8.97	8.10	7.53	5.02	4.55	3.46	3.29	3.48	4.61	5.28	7.48	8.16	(7)	
	HDD10.0	2327	450	395	325	167	59	7	1	1	27	142	305	449	(8)	
	HDD18.3	4661	708	628	582	408	264	140	46	60	182	383	552	707	(9)	
	CDD10.0	783	1	0	1	9	54	122	248	228	100	17	3	0	(10)	
	CDD18.3	77	0	0	0	2	6	36	29	5	0	0	0	0	(11)	
	CDH23.3	1516	0	0	0	2	43	107	613	583	163	4	0	0	(12)	
	CDH26.7	468	0	0	0	0	7	20	209	195	37	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.8	3.5	3.7	3.8	3.6	3.4	2.9	2.7	3.0	3.5	3.7	3.7	(14)	
(15) Precipitation	PrecAvg	548	27	24	34	49	76	107	47	46	47	33	31	28	(15)	
	PrecMax	844	43	49	64	94	166	323	173	118	135	79	61	59	(16)	
	PrecMin	343	8	6	12	17	9	24	3	3	5	6	10	9	(17)	
	PrecStd	120	9	11	13	17	38	67	38	28	29	19	13	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	13.6	16.4	22.2	27.2	29.5	33.5	33.2	30.4	23.2	17.5	12.2	(19)	
		MCWB	5.8	5.2	7.2	10.9	14.4	17.7	17.4	16.2	14.7	11.7	8.5	4.7	(20)	
	2%	DB	10.0	10.5	13.5	18.8	24.1	26.3	31.2	30.8	27.4	20.0	14.1	9.7	(21)	
		MCWB	4.2	4.3	6.3	9.1	12.7	15.3	16.9	15.8	13.7	10.3	7.2	4.0	(22)	
	5%	DB	8.1	8.2	11.4	16.0	21.6	24.0	29.1	28.9	25.1	17.0	11.7	7.7	(23)	
		MCWB	3.3	3.2	5.3	7.9	11.7	14.5	16.6	15.3	13.3	9.3	5.7	3.1	(24)	
	10%	DB	6.5	6.3	9.2	13.2	18.9	21.8	26.8	26.8	22.2	14.3	9.5	5.8	(25)	
		MCWB	2.5	1.9	3.9	6.4	10.5	13.2	16.0	15.0	12.3	7.7	4.4	2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.1	8.2	11.4	15.7	19.0	20.7	18.6	16.1	12.9	9.6	6.4	(27)	
		MCDB	11.9	11.9	14.8	20.6	25.4	25.7	29.0	26.6	26.2	21.4	15.8	10.7	(28)	
	2%	WB	5.0	4.8	6.9	9.7	13.5	16.7	19.1	17.5	14.9	11.1	7.6	4.6	(29)	
		MCDB	9.0	9.8	12.5	17.5	22.1	24.3	26.5	26.3	24.7	18.8	13.5	8.6	(30)	
	5%	WB	3.6	3.4	5.6	8.2	12.2	15.3	17.8	16.6	13.7	9.5	6.1	3.3	(31)	
		MCDB	7.7	8.1	10.9	15.1	20.2	22.1	25.8	25.6	23.0	16.2	11.2	7.1	(32)	
	10%	WB	2.5	1.9	4.1	6.8	11.0	14.0	16.8	15.7	12.6	8.1	4.6	2.0	(33)	
		MCDB	6.4	6.1	8.9	12.8	17.8	20.4	24.8	24.8	21.3	14.0	9.1	5.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.4	11.2	11.4	11.9	11.9	14.7	15.4	13.7	11.7	10.4	10.7	(35)	
		MCDBR	10.9	12.6	13.4	16.1	16.8	16.4	18.6	19.9	19.8	16.0	12.5	11.2	(36)	
	5% WB	MCWBR	7.2	8.1	7.9	8.4	8.0	8.0	7.9	8.0	8.8	8.5	7.3	7.4	(37)	
		MCWBR	10.9	12.6	12.9	15.0	15.2	14.7	16.4	17.1	17.9	15.2	12.0	11.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.243	0.257	0.276	0.322	0.330	0.326	0.332	0.348	0.311	0.279	0.255	0.231	(40)	
		taud	2.408	2.407	2.453	2.361	2.413	2.470	2.458	2.398	2.475	2.544	2.498	2.419	(41)	
	Ebn at Noon		843	913	948	927	929	932	921	889	895	877	826	814	(42)	
		Edh at Noon	62	79	91	112	111	106	105	107	89	69	57	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.24	2.24	3.47	4.79	5.70	6.38	7.11	5.68	4.04	2.50	1.39	0.99	(44)	
	RadStd		0.10	0.17	0.31	0.31	0.50	0.49	0.40	0.36	0.33	0.24	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page